NED University of Engineering and Technology, Karachi

Computer Science and Information Technology

COMPLEX COMPUTING PROBLEM (CCP)
PROJECT SUBMISSION

Knowledge-Based Systems (CT-359)

Project Title	Intelligent Knowledge-Based System for Industrial Equipment Fault Diagnosis and Maintenance Decision Support
Submission Date	

Student Details

Sr. No.	Student Name	Roll No.	Signature
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1. Project Instructions

- Complete all sections using the prescribed template. Replace bracketed placeholders with project-specific information.
- Submit the complete source code, dataset / knowledge base files, project report, and presentation according to the submission instructions.
- The implementation must be executable and demonstrable during the viva.
- Any assumptions made to complete the problem must be clearly stated and justified.
- All external datasets, knowledge sources, libraries, tools, or publications used in the project must be cited.
- The report and presentation must follow the prescribed template and be submitted on time.

2. Problem Statement

- Explain how a Knowledge-Based System can represent industrial equipment knowledge, including equipment, components, symptoms, faults, causes, and maintenance actions, using an appropriate knowledge representation formalism. (CLO 1)
- Analyze the given industrial equipment fault-diagnosis problem by identifying incomplete requirements, stakeholder needs, relationships, constraints, and suitable knowledge-based approaches for developing an effective solution. (CLO 2)
- Evaluate suitable knowledge representation methods and inference algorithms for diagnosing equipment faults and recommending maintenance actions, and justify the selected approach based on accuracy, explainability, safety, cost, and system requirements. (CLO 3)

3. Problem Description / Project Context

[Insert the finalized two-paragraph problem description here. Describe the industrial setting, heterogeneous knowledge sources, multiple symptoms and possible causes, stakeholder conflicts, incomplete information, and the need for diagnosis plus maintenance recommendation.]

4. Problem Analysis

4.1 Problem Understanding

[Enter your analysis here.]

4.2 Stakeholders and Their Requirements

[Enter your analysis here.]

4.3 Constraints and Conflicting Requirements

[Enter your analysis here.]

4.4 Root Cause / Requirement Identification

[Enter your analysis here.]

4.5 Assumptions

[Enter your analysis here.]

5. Knowledge Representation and KBS Design

5.1 Selected Knowledge Representation Formalism

[Enter details, diagrams, tables, rules, ontology/graph snapshots, or architecture here.]

5.2 Knowledge Base / Ontology / Rule Base

[Enter details, diagrams, tables, rules, ontology/graph snapshots, or architecture here.]

5.3 Inference Mechanism

[Enter details, diagrams, tables, rules, ontology/graph snapshots, or architecture here.]

5.4 Conflict Resolution Strategy

[Enter details, diagrams, tables, rules, ontology/graph snapshots, or architecture here.]

5.5 Explanation Facility
[Enter details, diagrams, tables, rules, ontology/graph snapshots, or architecture here.]

5.6 System Architecture
[Enter details, diagrams, tables, rules, ontology/graph snapshots, or architecture here.]

6. Dataset / Knowledge Sources

Source / Dataset	Type	Size / No. of Records	Purpose	Citation / URL
[Enter]	[Enter]	[Enter]	[Enter]	[Enter]

6. Implementation

6.1 Software / Tools / Libraries
[Enter details and include screenshots / code excerpts where required.]

6.2 Modules and Components
[Enter details and include screenshots / code excerpts where required.]

6.3 Code Structure
[Enter details and include screenshots / code excerpts where required.]

6.4 User Inputs and Outputs
[Enter details and include screenshots / code excerpts where required.]

7. Testing and Results

8. Evaluation, Limitations and Conclusion

8.1 Evaluation of the Proposed Approach
[Enter details.]

8.2 Limitations
[Enter details.]

8.3 Future Improvements
[Enter details.]

8.4 Conclusion
[Enter details.]

9. References